SPDI Structural Variation (SPDI-SV) Specification Draft (v0.4)

1. Overview

SPDI-SV extends SPDI for structural variant normalization aligned with VCF 4.5. It uses symbolic ALT and optional external sequence references.

2. Core Syntax

```
SEQ:POS:DEL_LEN:INS
```

```
SEQ:POS:DEL_LEN:INS|QUALIFIERS
```

3. Symbolic SV Representation

```
<INS>, <DEL>, <DUP>, <DUP:TANDEM>, <INV>, <CNV>
```

4. Sequence Handling

Insertion of Unknown or not provided sequence:

```
NC_000010.11:81000000:0:<INS>|SVLEN=2167
```

Insertion of Known external sequence:

```
NC_000010.11:81000000:0:<INS>|SVLEN=2167;SEQ_REF=NT_187361.1:45231-47397
```

5. Orientation

```
STRAND=REVERSE (forward is implicit)
```

6. SVLEN Definition

SVLEN is required for symbolic SVs and must be positive.

7. Event and Breakend Relationships

SPDI-SV uses VCF-compatible relational fields to express paired breakends and larger structural events.

MATEID identifies the corresponding mate variant. The value of MATEID MUST be the full SPDI-SV string of the mate variant.

If two variants identify each other via MATEID and are not part of a larger event, EVENT is not required.

If multiple variants participate in a larger or multi-break event, all related variants MUST include the same EVENT value.

EVENTTYPE SHOULD be used from the VCF controlled vocabulary to classify the event, where applicable.

```
Example (simple translocation):

NC_000001.11:1234567:1:G]NC_000005.10:7654321]|MATEID=NC_000005.10:7654321:1:T[NC_000001.11:1234567[

NC_000005.10:7654321:1:T[NC_000001.11:1234567[|MATEID=NC_000001.11:1234567:1:G]NC_000005.10:7654321]

Example (complex rearrangement):

NC_000001.11:30000000:1:A]NC_000002.11:40000000]|MATEID=NC_000002.11:40000000:1:T[NC_000001.11:30000000[;EVENT=evt1;EVENTTYPE=CHROMOTHRIPSIS

NC_000002.11:40000000:1:T[NC_000001.11:30000000[|MATEID=NC_000001.11:30000000:1:A]NC_000002.11:40000000];EVENT=evt1;EVENTTYPE=CHROMOTHRIPSIS
```

8. Examples by SV Types

Examples below use EVENT only for larger multi-break events. Paired breakends are linked using MATEID. Single-locus SVs do not include EVENT.

8.1 Copy Number Variation (CNV)

```
NC_000017.11:20000000:150000:<CNV>|SVLEN=150000;CN=3
```

Represents a copy number change across a single contiguous interval.

8.2 Insertion (INS)

```
NC_000010.11:81000000:0:<INS>|SVLEN=2167
```

```
NC_000010.11:81000000:0:<INS>|SVLEN=2167;SEQ_REF=NT_187361.1:45231-47397
```

Represents an insertion with unknown sequence or with external sequence reference.

8.3 Short Tandem Repeat (STR)

```
NC_000001.11:100000:0:<INS>|SVLEN=60
```

Represents repeat expansion as an insertion-length event.

8.4 Mobile Element Insertion (MEI)

```
NC_000006.12:32600000:0:<INS>|SVLEN=6000
```

Represents insertion of a mobile element.

8.5 Inversion (INV)

```
NC_000002.12:30000000:120000:<INV>|SVLEN=120000
```

Represents inversion of a genomic segment.

8.6 Deletion (DEL)

```
NC_000017.11:43045700:1500:<DEL>|SVLEN=1500
```

Represents deletion of reference bases.

8.7 Tandem Duplication (DUP:TANDEM)

```
NC_00008.12:25398284:0:<DUP:TANDEM>|SVLEN=40000
```

Represents duplication inserted adjacent to the original sequence.

8.8 Translocation / Breakend (BND)

```
NC_000001.11:1234567:1:G]NC_000005.10:7654321]|MATEID=NC_000005.10:7654321:1:T[NC_000001.11:1234567[
```

```
NC_000005.10:7654321:1:T[NC_000001.11:1234567[|MATEID=NC_000001.11:1234567:1:G]NC_000005.10:7654321]
```

Represents a paired breakend translocation; MATEID links the two records.

8.9 Delins (DELINS)

```
NC_000003.12:45000000:120:TTGCA
```

Represents a deletion replaced by an insertion at the same locus.

Appendix A: Glossary

Term	Definition
SPDI	Sequence Position Deletion Insertion model for variant normalization
SEQ	Reference sequence accession
POS	0-based interbase coordinate
DEL_LEN	Number of deleted bases
INS	Inserted sequence or symbolic ALT
SVLEN	Length of structural variant
SEQ_REF	External sequence reference accession:start-end
STRAND	Orientation; only REVERSE is explicitly specified
CN	Copy number
EVENT	Unique value tying together two or more records belonging to the same structural event
EVENTTYPE	Controlled vocabulary describing the type of structural event
MATEID	Identifier of the variant to which the current variant is joined; value is the full SPDI-SV string of the mate variant
DUP	Duplication
DUP:TANDEM	Tandem duplication
INV	Inversion
CNV	Copy number variation
BND	Breakend
DEL	Deletion
INS	Insertion

Appendix B: SV types in dbVar